

Welcome to Code Crunch!

C7 | CRUNCH LABS: Wire Crunch - Embedded Systems

TUE 1-3PM Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



linktr.ee/CODE.CRUNCH



Unit #0

C7 | Crunch LABS: Wire Crunch - Embedded Systems

Overview





Welcome to Code Crunch!

- Code Crunch is your go-to platform for collaboration, networking, and hands-on projects. We welcome freshmen, graduates, and alumni from any university, regardless of major or experience level.
- Join us virtually via Zoom or in person at Florida International University. Together, we'll break down barriers in technology, learn through immersive workshops, and build innovative solutions.



Our Mission & Vision

Mission:

Our mission is to equip students with the technical skills and professional connections needed to excel in coding interviews and beyond. We provide a platform where students can collaborate, learn, and grow professionally while emphasizing the importance of diversity and inclusion in the tech industry

Vision:

We strive to bridge the gap between academia and the tech industry by offering hands-on workshops, technical interview preparation, hackathons, and networking opportunities with experts from leading companies. Our vision is to empower students from all backgrounds to achieve their full potential, foster an inclusive tech community, and build a strong, supportive alumni network. By covering a wide range of technical areas, we aim to prepare students for success in a variety of tech careers.



What We Offer

- Hands-On Workshops: Topics include Python, JavaScript, React, and more.
- LeetCode Prep: Weekly sessions to master technical interviews.
- Hackathons: Collaborate on innovative projects in a competitive setting.
- Panel Discussions: Industry leaders from tech giants and startups share insights.
- Global Access: Code Crunch Global launches in October for worldwide participation!

Meet the E-Board

- **President:** Brian Bazurto
- **Vice President:** Amilcar Pena
- **Treasurer:** Bryan Garcia
- **Marketing/Logistics:** Henry Cruz
- **Logistics:** Jason Liang

Tech Leads:

- | | |
|--------------------|--------------------|
| • Alvisé Carpin | • Justin Joseph |
| • Arnold Miranda | • Mariana Phelan |
| • Daniel Hernandez | • Sara Salvatierra |
| • Gabriel Nogueras | • Thierry Laguerre |





Why Join Code Crunch?

- Collaborate with peers and mentors.
- Develop technical and leadership skills.
- Network with professionals from top companies.
- Build impactful projects and gain hands-on experience.

A welcoming and inclusive space for underrepresented groups in tech.



Spring 2025 Weekly Schedule



Funded by SGA | Open to All Students

CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Cost: No-cost January 6, 2025 - April 4, 2025

Your CS Club for Collaboration, Networking, and Hands-On Projects!
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

Crunch AI Data Science Python MON 1 PM - 2 PM WED 1 PM - 2 PM	CodeCrunch LABS Portfolio Java & React MON 2 PM - 3 PM	Wire Crunch: Embedded Systems C++ TUE 1 PM - 2 PM	Crunch Convos Python WED 2 PM - 3 PM	Cybersecurity Crunch VM/LINUX THU 1 PM - 2 PM
CodeCrunch LABS Hackathon Java & React THU 2 PM - 3 PM	CodeCrunch LABS: Web Dev Javascript THU 3 PM - 4 PM	Crunch The Code Python/ Leetcode FRI 1 PM - 2 PM	Swe Convos Networking FRI 2 PM - 3 PM	

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on
[LU.MA/CODECRUNCH](https://lu.ma/codecrunch)

In person or
Stream on
Zoom

 [GO.FIU.EDU/CODECRUNCH](https://go.fiu.edu/codecrunch)

 [LINKTR.EE/CODE.CRUNCH](https://linktr.ee/code.crunch)



Become a member





Be Part of the Code Crunch Community

1. RSVP for sessions at go.fiu.edu/codecrunch.
2. Follow us on LinkedIn, Instagram, and Discord for updates.
3. Become a member using the QR code on this slide.
4. Attend workshops, participate in events, and join our leadership team.

Together, we innovate and grow.



Welcome to the course!

Weekly Overview:

- Week 1: Introduction to embedded systems
- Week 2: Unit Testing
- Week 3: Sensors and Data acquisition
- Week 4: Serial Communication
- Week 5: Timers and Interrupts
- Week 6: Pulse Width Modulation and Unit Testing
- Week 7: Intro to I2C and SPI Communication
- Week 8: Wireless Communication
- Week 9: Real Time Systems
- Week 10: Recap Project

What are Embedded Systems?



- A combination of hardware and software designed for a specific task

Pros:

- Efficient
- Real-time operation
- Low cost

Const:

- Limited adaptability
- Limited Resources
- Attack Replicaiton

The Importance of Pins

- Pins help us communicate with the microcontroller
 - Establishes a connection between devices
 - Color matters when connecting cable to the pin
- The connection depends on what type of communication it has with the board
 - Ex. An analog switch gets plugged into AO pin



Resources Matter!

- Since there is limited space, there are limited resources
- We often find ourselves optimizing code to improve performance and decrease latency/overheads

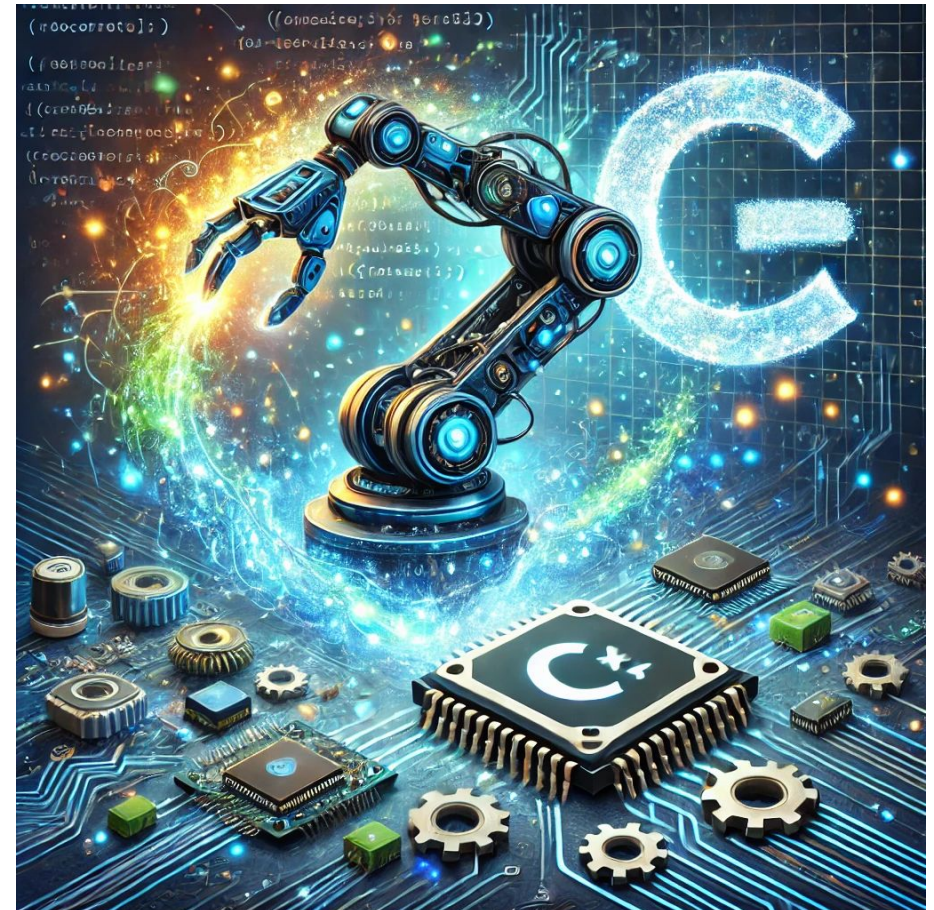
Tips:

- Avoid using dynamic space
 - Ex. new or malloc
- use volatile depending on how a variable is accessed
 - Ex. variables in an interrupt or shared resources



Why C++?

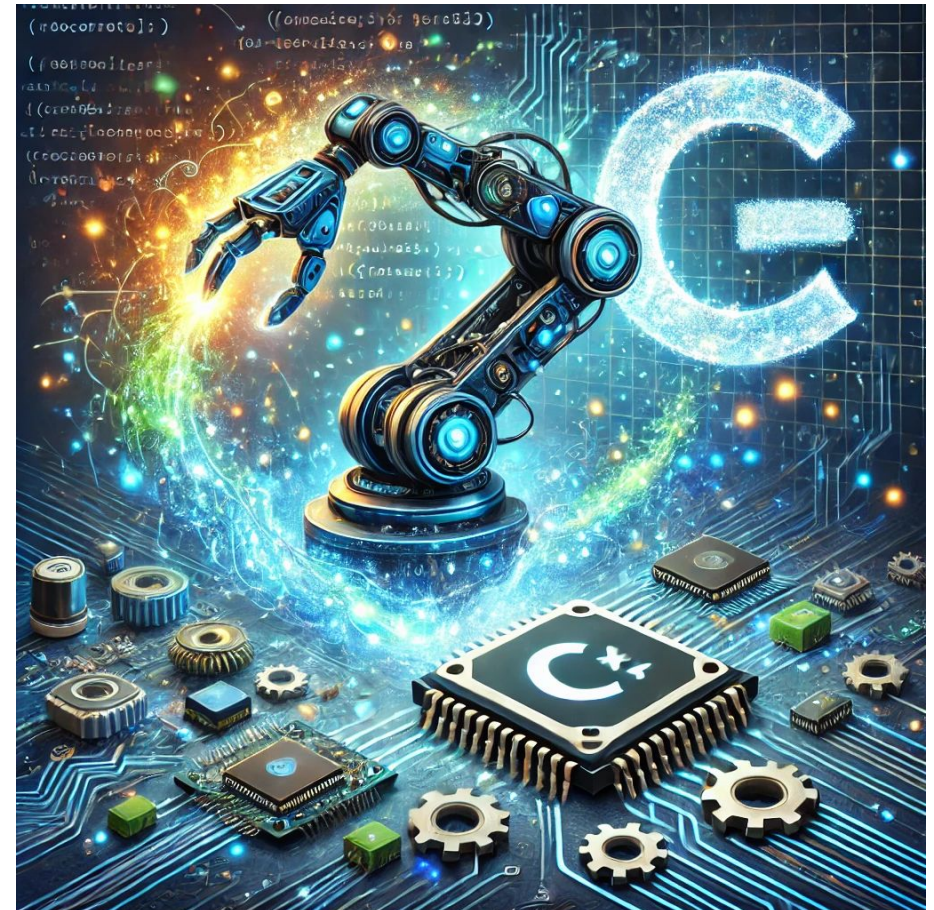
- Direct communication with CPU
- High performance
- Low level hardware access
- Low resource usage
- Real time capabilities
- Portability



Arduino Software Basics



- setup function
 - Where we initialize the state of the pins we are using (Inputs/Outputs)
- Loop function
 - This runs the body of code repeatedly until board is disconnected



Digital Vs. Analog Signals



Digital Signals:

- Digital states like HIGH and LOW
- turns LEDs on and off

Analog Signals:

- Continuous range of values
- Light sensors, moisture sensors

Digital Communicating on an Arduino Board



- Arduino has pre-existing functions for digital and analog
 - `pinMode(pin,I/O)` -> configures a pin as input or output
 - Digital Input
 - `digitalWrite(pin,HIGH/LOW)` -> sets a digital output pin state
 - Digital Output
 - `digitalRead(pin)` -> Reads a digital input state

Analog Communicating on an Arduino Board



- Analog Input
 - `analogRead(pin)` reads voltages (0-5v) and returns a value of 0-1023
- Analog Output
 - `analogWrite(pin, value)` for PWM-enabled pins
 - This allows us to dim lights/control electricity flow



Q&A Session



Let's Innovate Together!

Thank you for joining us on this exciting journey! Code Crunch is redefining tech education and fostering inclusivity in technology. Let's collaborate, innovate, and create a brighter future together.

RSVP for sessions today. Follow us on social media. Spread the word about Code Crunch!

#CodeCrunch #CodeCrunchFIU #DiversityInTech #InclusiveTech #LeadershipOpportunities
#FIUTechCommunity #WomenInSTEM #LGBTQInTech #LatinxInTech #BlackInTech
#AAPIsInTech #TechLeadership #JoinOurTeam #FIUInnovation #TechForEveryone
#EmpowerTech #FutureLeaders #HackerCulture #CodeYourFuture #TechDiversity
#FIUPanthers #STEMOpportunities #BuildYourFuture #LeadWithCode #PanthersInTech



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



RSVP lu.ma/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch